

Babu Banarasi Das University



NO SQL Assignment

SUBMITTED TO
Mr Harendra Singh

SUBMITTED BY:
Name:Aditi Tiwari

Roll No: 32

Class: BCADS21

1. CREATE DATABASE

use Inventory

```
test> use Inventory
switched to db Inventory
Inventory> █
```

2. CREATE COLLECTION

db.createCollection("products")

```
Inventory> db.createCollection("products")
{ ok: 1 }
Inventory> █
```

3. INSERT 50 PRODUCTS

```
db.products.insertMany([
  { productId: 1, name: "Wireless Mouse", category: "Electronics", price: 880, warehouse: "Lucknow" },
  { productId: 2, name: "Basmati Rice 5kg", category: "Grocery", price: 760, warehouse: "Jaunpur" },
  { productId: 3, name: "Bluetooth Speaker", category: "Electronics", price: 920, warehouse: "Kanpur" },
  { productId: 4, name: "Sunflower Oil 1L", category: "Grocery", price: 690, warehouse: "Ayodhya" },
  { productId: 5, name: "LED Monitor", category: "Electronics", price: 810, warehouse: "Varanasi" },
  { productId: 6, name: "Office Chair", category: "Furniture", price: 850, warehouse: "Lucknow" },
  { productId: 7, name: "Smartphone Charger", category: "Electronics", price: 730, warehouse: "Prayagraj" },
  { productId: 8, name: "Laptop Stand", category: "Electronics", price: 900, warehouse: "Lucknow" },
  { productId: 9, name: "Wooden Bookshelf", category: "Furniture", price: 780, warehouse: "Gorakhpur" },
  { productId: 10, name: "Wheat Flour 5kg", category: "Grocery", price: 650, warehouse: "Kanpur" },
  { productId: 11, name: "Study Table", category: "Furniture", price: 710, warehouse: "Jaunpur" },
  { productId: 12, name: "USB Hub", category: "Electronics", price: 950, warehouse: "Lucknow" },
  { productId: 13, name: "Green Tea Pack", category: "Grocery", price: 840, warehouse: "Ayodhya" },
  { productId: 14, name: "Smartwatch", category: "Electronics", price: 870, warehouse: "Prayagraj" },
  { productId: 15, name: "Dining Table", category: "Furniture", price: 740, warehouse: "Varanasi" },
  { productId: 16, name: "Organic Honey", category: "Grocery", price: 890, warehouse: "Lucknow" },
  { productId: 17, name: "Power Bank", category: "Electronics", price: 820, warehouse: "Kanpur" },
  { productId: 18, name: "Masala Tea", category: "Grocery", price: 580, warehouse: "Gonda" },
  { productId: 19, name: "Sofa Set", category: "Furniture", price: 910, warehouse: "Lucknow" },
  { productId: 20, name: "Wireless Earbuds", category: "Electronics", price: 630, warehouse: "Bareilly" },
  { productId: 21, name: "Peanut Butter Jar", category: "Grocery", price: 770, warehouse: "Lucknow" },
  { productId: 22, name: "Webcam", category: "Electronics", price: 860, warehouse: "Jaunpur" },
  { productId: 23, name: "Bed Frame", category: "Furniture", price: 720, warehouse: "Kanpur" },
  { productId: 24, name: "WiFi Router", category: "Electronics", price: 800, warehouse: "Ayodhya" },
  { productId: 25, name: "Cornflakes Box", category: "Grocery", price: 670, warehouse: "Prayagraj" },
  { productId: 26, name: "Wardrobe", category: "Furniture", price: 940, warehouse: "Lucknow" },
  { productId: 27, name: "Mechanical Keyboard", category: "Electronics", price: 880, warehouse: "Varanasi" },
  { productId: 28, name: "Almonds Pack", category: "Grocery", price: 610, warehouse: "Gorakhpur" },
  { productId: 29, name: "Tablet Cover", category: "Electronics", price: 750, warehouse: "Kanpur" },
  { productId: 30, name: "Coffee Table", category: "Furniture", price: 790, warehouse: "Lucknow" },
  { productId: 31, name: "Pasta Pack", category: "Grocery", price: 680, warehouse: "Jaunpur" },
  { productId: 32, name: "Extension Board", category: "Electronics", price: 900, warehouse: "Lucknow" },
  { productId: 33, name: "Recliner Chair", category: "Furniture", price: 550, warehouse: "Ayodhya" },
  { productId: 34, name: "Olive Oil", category: "Grocery", price: 830, warehouse: "Kanpur" },
  { productId: 35, name: "Cooling Fan", category: "Electronics", price: 960, warehouse: "Prayagraj" },
  { productId: 36, name: "Shoe Rack", category: "Furniture", price: 700, warehouse: "Lucknow" },
  { productId: 37, name: "Coffee Beans", category: "Grocery", price: 810, warehouse: "Varanasi" },
  { productId: 38, name: "Pen Drive", category: "Electronics", price: 660, warehouse: "Gorakhpur" },
  { productId: 39, name: "TV Unit", category: "Furniture", price: 930, warehouse: "Lucknow" },
  { productId: 40, name: "Graphics Card", category: "Electronics", price: 850, warehouse: "Kanpur" },
  { productId: 41, name: "Mixed Spices Pack", category: "Grocery", price: 740, warehouse: "Jaunpur" },
  { productId: 42, name: "Dressing Table", category: "Furniture", price: 890, warehouse: "Lucknow" },
  { productId: 43, name: "Smart Bulb", category: "Electronics", price: 620, warehouse: "Ayodhya" },
  { productId: 44, name: "Dry Fruits Box", category: "Grocery", price: 780, warehouse: "Prayagraj" },
  { productId: 45, name: "Bunk Bed", category: "Furniture", price: 840, warehouse: "Lucknow" },
  { productId: 46, name: "HDMI Cable", category: "Electronics", price: 730, warehouse: "Kanpur" },
  { productId: 47, name: "Sugar 1kg", category: "Grocery", price: 590, warehouse: "Gonda" },
  { productId: 48, name: "Filing Cabinet", category: "Furniture", price: 870, warehouse: "Lucknow" },
  { productId: 49, name: "Portable SSD", category: "Electronics", price: 640, warehouse: "Bareilly" },
  { productId: 50, name: "Salt 1kg", category: "Grocery", price: 910, warehouse: "Varanasi" }
])
```

```
'19': ObjectId('6ab7df10fa70e9ba4d075e0a'),
'20': ObjectId('6ab7df10fa70e9ba4d075e0b'),
'21': ObjectId('6ab7df10fa70e9ba4d075e0c'),
'22': ObjectId('6ab7df10fa70e9ba4d075e0d'),
'23': ObjectId('6ab7df10fa70e9ba4d075e0e'),
'24': ObjectId('6ab7df10fa70e9ba4d075e0f'),
'25': ObjectId('6ab7df10fa70e9ba4d075e10'),
'26': ObjectId('6ab7df10fa70e9ba4d075e11'),
'27': ObjectId('6ab7df10fa70e9ba4d075e12'),
'28': ObjectId('6ab7df10fa70e9ba4d075e13'),
'29': ObjectId('6ab7df10fa70e9ba4d075e14'),
'30': ObjectId('6ab7df10fa70e9ba4d075e15'),
'31': ObjectId('6ab7df10fa70e9ba4d075e16'),
'32': ObjectId('6ab7df10fa70e9ba4d075e17'),
'33': ObjectId('6ab7df10fa70e9ba4d075e18'),
'34': ObjectId('6ab7df10fa70e9ba4d075e19'),
'35': ObjectId('6ab7df10fa70e9ba4d075e1a'),
'36': ObjectId('6ab7df10fa70e9ba4d075e1b'),
'37': ObjectId('6ab7df10fa70e9ba4d075e1c'),
'38': ObjectId('6ab7df10fa70e9ba4d075e1d'),
'39': ObjectId('6ab7df10fa70e9ba4d075e1e'),
'40': ObjectId('6ab7df10fa70e9ba4d075e1f'),
'41': ObjectId('6ab7df10fa70e9ba4d075e20'),
'42': ObjectId('6ab7df10fa70e9ba4d075e21'),
'43': ObjectId('6ab7df10fa70e9ba4d075e22'),
'44': ObjectId('6ab7df10fa70e9ba4d075e23'),
'45': ObjectId('6ab7df10fa70e9ba4d075e24'),
'46': ObjectId('6ab7df10fa70e9ba4d075e25'),
'47': ObjectId('6ab7df10fa70e9ba4d075e26'),
'48': ObjectId('6ab7df10fa70e9ba4d075e27'),
'49': ObjectId('6ab7df10fa70e9ba4d075e28')
}
}
Inventory> █
```

4. DISPLAY ALL PRODUCTS

```
db.products.find()
```

```
Inventory> db.products.find()
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df8'),
  productId: 2,
  name: 'Basmati Rice 5kg',
  category: 'Grocery',
  price: 760,
  warehouse: 'Jaunpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfa'),
  productId: 4,
  name: 'Sunflower Oil 1L',
  category: 'Grocery',
  price: 690,
  warehouse: 'Ayodhya'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfb'),
  productId: 5,
  name: 'Smart Watch',
  category: 'Electronics',
  price: 1200,
  warehouse: 'Lucknow'
}
```

5. EQUAL TO — \$eq

```
db.products.find({ price: { $eq: 880 } })
```

```
Inventory> db.products.find({ price: { $eq: 880 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e11'),
  productId: 27,
  name: 'Mechanical Keyboard',
  category: 'Electronics',
  price: 880,
  warehouse: 'Varanasi'
}
Inventory> █
```

6. GREATER THAN — \$gt

```
db.products.find({ price: { $gt: 850 } })
```

```
Inventory> db.products.find({ price: { $gt: 850 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',
  category: 'Electronics',
  price: 900,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e02'),
  productId: 12,
  name: 'USB Hub',
  category: 'Electronics',
  price: 950,
  warehouse: 'Lucknow'
}
{
```

7. LESS THAN — \$lt

```
db.products.find({ price: { $lt: 600 } })
```

```
Inventory> db.products.find({ price: { $lt: 600 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e08'),
  productId: 18,
  name: 'Masala Tea',
  category: 'Grocery',
  price: 580,
  warehouse: 'Gonda'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e17'),
  productId: 33,
  name: 'Recliner Chair',
  category: 'Furniture',
  price: 550,
  warehouse: 'Ayodhya'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e25'),
  productId: 47,
  name: 'Sugar 1kg',
  category: 'Grocery',
  price: 590,
  warehouse: 'Gonda'
}
Inventory> █
```

8. GREATER THAN OR EQUAL TO — \$gte

```
db.products.find({ price: { $gte: 900 } })
```

```
Inventory> db.products.find({ price: { $gte: 900 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',
  category: 'Electronics',
  price: 900,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e02'),
  productId: 12,
  name: 'USB Hub',
  category: 'Electronics',
  price: 950,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e09'),
  productId: 19,
  name: 'Sofa Set',
  category: 'Furniture',
  price: 910,
  warehouse: 'Lucknow'
}
{
```

9. LESS THAN OR EQUAL TO — \$lte

```
db.products.find({ price: { $lte: 700 } })
```

```
Inventory> db.products.find({ price: { $lte: 700 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfa'),
  productId: 4,
  name: 'Sunflower Oil 1L',
  category: 'Grocery',
  price: 690,
  warehouse: 'Ayodhya'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e00'),
  productId: 10,
  name: 'Wheat Flour 5kg',
  category: 'Grocery',
  price: 650,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e08'),
  productId: 18,
  name: 'Masala Tea',
  category: 'Grocery',
  price: 580,
  warehouse: 'Gonda'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e0a'),
  productId: 20,
  name: 'Wireless Earbuds',
  category: 'Electronics',
  price: 630,
  warehouse: 'Bareilly'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e0c'),
  productId: 22,
  name: 'Smart Watch',
  category: 'Electronics',
  price: 750,
  warehouse: 'Bareilly'
}
```

10. UPDATE OPERATION

```
db.products.updateOne(
  { productId: 5 },
  { $set: { price: 950 } }
)
```

```
Inventory> db.products.updateOne(
...   { productId: 5 },
...   { $set: { price: 950 } }
... )
... {
...   acknowledged: true,
...   matchedCount: 1,
...   modifiedCount: 1
... }
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Inventory> █
```

11. DELETE ONE OPERATION

```
db.products.deleteOne({ productId: 10 })
```

```
Inventory> db.products.deleteOne({ productId: 10 })
{ acknowledged: true, deletedCount: 1 }
Inventory> █
```

12. DELETE MANY OPERATION

```
db.products.deleteMany({
  price: { $lt: 600 }
})
```

```
Inventory> db.products.deleteMany({
...   price: { $lt: 600 }
... })
... { acknowledged: true, deletedCount: 3 }
{ acknowledged: true, deletedCount: 3 }
Inventory> █
```

13. AND OPERATION — \$and

```
db.products.find({
  $and: [
    { price: { $gt: 800 } },
    { category: "Electronics" }
  ]
})
```

```
Inventory> db.products.find({
...   $and: [
...     { price: { $gt: 800 } },
...     { category: "Electronics" }
...   ]
... })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfb'),
  productId: 5,
  name: 'LED Monitor',
  category: 'Electronics',
  price: 950,
  warehouse: 'Varanasi'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',

```

14. OR OPERATION — \$or

```
db.products.find({
  $or: [
    { warehouse: "Lucknow" },
    { warehouse: "Kanpur" }
  ]
})
```

```
Inventory> db.products.find({
...   $or: [
...     { warehouse: "Lucknow" },
...     { warehouse: "Kanpur" }
...   ]
... })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfc'),
  productId: 6,
  name: 'Office Chair',
  category: 'Furniture',
  price: 850,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',
```

15. SORT OPERATION

A. Ascending

```
db.products.find().sort({ price: 1 })
```

```
Inventory> db.products.find().sort({ price: 1 })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e12'),
  productId: 28,
  name: 'Almonds Pack',
  category: 'Grocery',
  price: 610,
  warehouse: 'Gorakhpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e21'),
  productId: 43,
  name: 'Smart Bulb',
  category: 'Electronics',
  price: 620,
  warehouse: 'Ayodhya'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e0a'),
  productId: 20,
  name: 'Wireless Earbuds',
  category: 'Electronics',
  price: 630,
  warehouse: 'Bareilly'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e27'),
  productId: 49,
  name: 'Portable SSD',
  category: 'Electronics',
  price: 640,
  warehouse: 'Bareilly'
}
{
```

B. Descending

```
db.products.find().sort({ price: -1 })
```

```
Inventory> db.products.find().sort({ price: -1 })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e19'),
  productId: 35,
  name: 'Cooling Fan',
  category: 'Electronics',
  price: 960,
  warehouse: 'Prayagraj'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfb'),
  productId: 5,
  name: 'LED Monitor',
  category: 'Electronics',
  price: 950,
  warehouse: 'Varanasi'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e02'),
  productId: 12,
  name: 'USB Hub',
  category: 'Electronics',
  price: 950,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e10'),
  productId: 26,
  name: 'Wardrobe',
  category: 'Furniture',
  price: 940,
  warehouse: 'Lucknow'
}
{
```

16. LIMIT OPERATION

```
db.products.find().limit(5)
```

```
Inventory> db.products.find().limit(5)
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df8'),
  productId: 2,
  name: 'Basmati Rice 5kg',
  category: 'Grocery',
  price: 760,
  warehouse: 'Jaunpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfa'),
  productId: 4,
  name: 'Sunflower Oil 1L',
  category: 'Grocery',
  price: 690,
  warehouse: 'Ayodhya'
}
{
```

17. SEARCH BY WAREHOUSE

```
db.products.find({ warehouse: "Lucknow" })
```

```
Inventory> db.products.find({ warehouse: "Lucknow" })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfc'),
  productId: 6,
  name: 'Office Chair',
  category: 'Furniture',
  price: 850,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',
  category: 'Electronics',
  price: 900,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e02'),
  productId: 12,
  name: 'USB Hub',
  category: 'Electronics',
  price: 950,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e03'),
  productId: 13,
  name: 'Wireless Keyboard',
  category: 'Electronics',
  price: 1200,
  warehouse: 'Lucknow'
}
```

18. COUNT DOCUMENTS

```
db.products.countDocuments()
```

```
Inventory> db.products.countDocuments()
46
Inventory> █
```